

# FOOD DELIVERY ORDER TRACKING SYSTEM

## A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

**Capstone Project** to the award of the degree of

**BACHELOR OF TECHNOLOGY**

IN

**INFORMATION TECHNOLOGY**

**Submitted by**

**K.DINESH(1921260051)**

**Under the Supervision of Dr. K. JAISHARMA**

**SIMATS ENGINEERING**

Saveetha Institute of Medical and Technical Sciences Chennai-602105

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**SIMATS ENGINEERING**

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

## BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“Food Delivery Order Tracking System”** has been carried out by K.DINESH **(1921260051)** under the supervision of **Dr. K. Jaisharma** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Information Technology program at Saveetha Institute of Medical and Technical Sciences, Chennai.

| COURSE COORDINATOR                                      | COURSE FACULTY / GUIDE                                                   |
|---------------------------------------------------------|--------------------------------------------------------------------------|
| Dr.S.SIVAGAMI<br>Professor<br>Cse<br>SIMATS ENGINEERING | Dr.K.JAISHARMA<br>Professor<br>Cse/neural networks<br>SIMATS ENGINEERING |

## DECLARATION

We, K. DINESH (1921260051) of the Department of

Information Technology, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences,

Chennai, hereby declare that the Capstone Project Work entitled **“Food Delivery Order Tracking System”** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate and carried out in accordance with engineering ethics and academic integrity. All sources, references, data, software and other materials used in the project will be appropriately acknowledged.

We further declare that the data, results, analysis and findings presented in this report will not be fabricated, falsified or manipulated. Any external tools, software, datasets, computational resources or AI-assisted tools used in the project will be appropriately disclosed.

Place: Chennai

Date: September 2026

Signatures of the Students with Names

## INDIVIDUAL CONTRIBUTION STATEMENT

| Student                 | Responsibilities                | Design & Development              | Testing & Analysis         | Report                           | %     |
|-------------------------|---------------------------------|-----------------------------------|----------------------------|----------------------------------|-------|
| Dinesh<br>1921260058    | Backend and database support    | Order and status modules          | Functional testing         | Implementation sections          | 33.3% |
| Kiruthika<br>1924260011 | Frontend and user interface     | Customer tracking screens         | UI and integration testing | Testing/analysis                 | 33.3% |
| Akshaya<br>1921260087   | System design and documentation | Tracking workflow and integration | System-level evaluation    | Report coordination/presentation | 33.4% |
| Total                   | —                               | —                                 | —                          | —                                | 100%  |

## ABSTRACT

Food delivery services depend on accurate order processing and timely communication between customers, restaurants and delivery personnel. A common difficulty is that customers may not know the current status of an order after placing it. The Food Delivery Order Tracking System is proposed as a software-based solution that allows users to place or view orders, track order progress and receive clear status information through a centralized application.

The system is designed around an order lifecycle such as Order Placed, Restaurant Accepted, Food Preparing, Ready for Pickup, Picked Up, Out for Delivery and Delivered. The application stores order details and status changes so that customers and authorized staff can view the latest state of an order. A modular design separates user interface, order management, tracking logic and data storage.

The project methodology includes requirement analysis, system design, database modelling, interface development, implementation, functional testing and evaluation. The prototype is intended as an academic system and does not claim production-scale performance, real-world GPS accuracy or commercial payment-security certification. Actual measured performance should be inserted after testing.

**Keywords:** Food Delivery, Order Tracking, Web Application, Database, Status Management, Real-Time Tracking

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# LIST OF ABBREVIATIONS / SYMBOLS

| Symbol | Description                       | Unit |
|--------|-----------------------------------|------|
| UI     | User Interface                    | —    |
| DB     | Database                          | —    |
| API    | Application Programming Interface | —    |
| GPS    | Global Positioning System         | —    |
| ETA    | Estimated Time of Arrival         | min  |
| CRUD   | Create, Read, Update, Delete      | —    |

# CHAPTER 1 INTRODUCTION AND ENGINEERING PROBLEM

## 1.1 Background

Online food delivery involves several connected activities: customer ordering, restaurant acceptance, food preparation, pickup and delivery. When these stages are not communicated clearly, customers may be uncertain about whether the order has been accepted, prepared or dispatched. A digital tracking system provides a structured way to store and communicate order status.

## 1.2 Need for the Project

Customers need transparent order information after placing an order. Restaurants need an organized order-status workflow, while delivery personnel need to know which orders are ready for pickup and delivery. A tracking system reduces uncertainty by maintaining a common status record and presenting the latest available state to authorized users.

## 1.3 Problem Statement

Design and develop a food delivery order tracking system that records customer orders and provides a clear, sequential view of order status from placement to delivery, while maintaining data consistency, usability and role-based access.

## 1.4 Project Aim

To develop a user-friendly software system for managing and tracking food delivery orders through a centralized order-status workflow.

## 1.5 Project Objectives

1. To identify requirements of customers, restaurants and delivery staff.
2. To design an order-status workflow and database structure.
3. To develop interfaces for order viewing and tracking.
4. To implement status updates across the order lifecycle.
5. To test core functions, data validation and status transitions.
6. To evaluate system strengths, limitations and future improvements.

## 1.6 Scope

**Included:** user/order records, order placement representation, order-status management, tracking interface, database storage, basic role separation and functional testing.

**Excluded:** production payment processing, certified cybersecurity, live commercial GPS infrastructure, restaurant POS integration and large-scale deployment.

**Assumptions:** users have access to a supported device and network where required; authorized staff update statuses correctly; database connectivity is available.

**Boundary conditions:** the academic prototype demonstrates order tracking logic and interface behaviour and should not be treated as a production food-delivery platform.

## 1.7 Expected Engineering Outcome

The intended outcome is a working software prototype in which an order is stored with a unique identifier and its status progresses through defined stages. The customer can view the latest status, while authorized system users can update the order according to their role.

## Engineering Problem Summary

The central engineering challenge is maintaining consistent order state across multiple actors. The design must prevent invalid status transitions, present information clearly, preserve order history where required and remain simple enough for an academic prototype.

# CHAPTER 2 LITERATURE REVIEW AND EXISTING SOLUTIONS

## 2.1 Review Method

The review considers the project requirements and established software concepts relevant to online ordering, transaction workflows, database systems, role-based access and status tracking. Existing approaches are compared according to their suitability for a student-level order tracking prototype.

## 2.2 Review of Existing Work

Traditional telephone-based ordering provides limited visibility after an order is placed. Basic online ordering systems improve order submission but may provide only a small amount of status information. Modern delivery platforms commonly combine ordering, restaurant processing, courier assignment and customer tracking. Such systems can provide rich real-time information but require complex infrastructure and integrations.

## 2.3 Comparative Analysis

| Existing Approach      | Advantages                      | Limitations                 | Relevance              |
|------------------------|---------------------------------|-----------------------------|------------------------|
| Phone/manual ordering  | Simple, low infrastructure      | Poor status visibility      | Baseline               |
| Basic online ordering  | Digital order capture           | Limited tracking            | Relevant               |
| Full delivery platform | Rich tracking and integrations  | High complexity/cost        | Future-scale reference |
| Proposed prototype     | Clear workflow, simple tracking | Limited production features | Directly relevant      |

## 2.4 Research / Engineering Gap

A student project needs to demonstrate the core engineering principle of order tracking without reproducing the infrastructure of a commercial delivery company. The gap addressed by this project is therefore the creation of a focused, understandable prototype that demonstrates order-state management, user visibility and database-backed tracking.

## 2.5 Proposed Contribution

The proposed system contributes a modular order tracking workflow connecting user interaction, order records, status transitions and tracking display. The architecture can later be extended with maps, live location, notifications, payment integration and analytics.

## Literature Review Conclusion

The review supports a database-backed, status-driven architecture as an appropriate foundation. Realtime GPS tracking is treated as an optional extension because it introduces additional hardware, network and privacy requirements beyond the core academic objective.

# CHAPTER 3 ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

## 3.1 Requirements

| Stakeholder    | Requirement                                           |
|----------------|-------------------------------------------------------|
| Customer       | Place/view order and see current order status         |
| Restaurant     | View incoming orders and update preparation status    |
| Delivery staff | View assigned order and update pickup/delivery status |
| Administrator  | Manage users, orders and system records               |
| Type           | Measurable Target                                     |
| Functional     | Create and retrieve order records                     |
| Functional     | Allow valid status updates                            |
| Functional     | Display latest status to customer                     |
| Non-functional | Clear, consistent and responsive interface            |
| Non-functional | Role-based access for protected actions               |

## 3.2 Constraints & Applicable Standards

Constraints include limited development time, database reliability, device/browser compatibility, authentication design, data privacy and project budget. The final implementation should follow the security and privacy requirements applicable to the chosen development stack and institutional guidelines. Any production deployment would require additional security assessment.

## 3.3 Design Specifications

| Parameter | Target / Requirement              | Unit | Priority |
|-----------|-----------------------------------|------|----------|
| Order ID  | Unique identifier for every order | —    | High     |

|                  |                                   |   |      |
|------------------|-----------------------------------|---|------|
| Status           | Defined valid lifecycle stages    | — | High |
| Tracking display | Shows current order state clearly | — | High |
| Data storage     | Persistent order/customer records | — | High |
| Access control   | Protected staff actions           | — | High |

### 3.4 Alternative Solutions & Evaluation

| Criterion             | Simple Static App | Database Web App | Full Real-Time Platform |
|-----------------------|-------------------|------------------|-------------------------|
| Performance           | Medium            | Good             | High                    |
| Cost                  | Low               | Low–Medium       | High                    |
| Implementation effort | Low               | Medium           | Very high               |
| Tracking capability   | Limited           | Good             | Excellent               |
| Scalability           | Low               | Medium           | High                    |

The database-backed web application is selected because it demonstrates meaningful order tracking while remaining feasible for a student capstone.

### 3.5 Selected Design & Detailed Design

**Architecture:** User Interface → Application/API Layer → Authentication & Order Logic → Database → Tracking Status Display.

**Order flow:** Customer places order → Order record created → Restaurant accepts → Food preparing → Ready for pickup → Delivery staff picks up → Out for delivery → Delivered.

The database stores users, restaurants, orders, order items and status information. The application layer validates requests and controls which roles can change particular states. The user interface presents the current status in a simple sequence so that the customer can understand progress.

A normalized database structure reduces duplicate information. A unique order identifier provides a stable reference for retrieval. Status-transition validation is important because an order should not normally move directly from Order Placed to Delivered without the required intermediate actions.

3.6 Trade-off Analysis

| Decision              | Alternative        | Benefit                           | Disadvantage               | Final Decision                  |
|-----------------------|--------------------|-----------------------------------|----------------------------|---------------------------------|
| Status-based tracking | Live GPS first     | Simple and reliable for prototype | No map location            | Selected for core project       |
| Database storage      | In-memory storage  | Persistent records                | Requires setup             | Selected for realistic workflow |
| Role-based access     | Single-user access | Better control                    | More implementation effort | Selected                        |

3.7 Sustainability, Ethics, Safety, Risk & Security

| Domain         | Consideration                                     | Evidence Evaluated                      | Impact on Decision                       |
|----------------|---------------------------------------------------|-----------------------------------------|------------------------------------------|
| Sustainability | Software reduces paper-based status communication | Digital workflow design                 | Use electronic records where appropriate |
| Ethics         | Avoid misleading delivery estimates               | Status/ETA limitations                  | Show only supported status information   |
| Privacy        | Protect customer/order information                | Database fields and access roles        | Restrict protected records               |
| Security       | Prevent unauthorized order changes                | Authentication and authorization design | Role-based protected operations          |

Risk Register

| Risk                      | Likelihood | Severity | Mitigation                           | Residual   |
|---------------------------|------------|----------|--------------------------------------|------------|
| Invalid status transition | Medium     | High     | Validate allowed transitions         | Low        |
| Unauthorized update       | Medium     | High     | Authentication/role checks           | Low–Medium |
| Database failure          | Low–Medium | High     | Backups and validation               | Low–Medium |
| Incorrect ETA             | Medium     | Medium   | Treat ETA as estimate, not guarantee | Low        |

# CHAPTER 4 IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

## 4.1 Development Approach & Resources

The project follows an iterative software-development approach: requirements are identified, the system architecture and database are designed, interfaces are developed, order-status logic is implemented, modules are integrated and then tested. Development resources may include a code editor/IDE, browser, database system, version-control repository and local or hosted application environment.

## 4.2 Software Implementation & Modern Tools Used

| Tool / Software          | Purpose                                | Stage Used     |
|--------------------------|----------------------------------------|----------------|
| Code editor / IDE        | Write and manage application code      | Development    |
| Web browser              | Run and test user interface            | Testing        |
| Database system          | Store users, orders and status records | Implementation |
| Version control          | Maintain source changes                | Development    |
| API/testing tool if used | Test application endpoints             | Verification   |

Implementation modules include user authentication, order management, restaurant/order processing, delivery-status management, customer tracking and database operations. Insert actual screenshots of the completed interface and database here in the final submission.

## 4.3 Test Plan & Verification

| Requirement               | Test Method                        | Acceptance Criterion         | Achieved     | Status       |
|---------------------------|------------------------------------|------------------------------|--------------|--------------|
| Create order              | Submit valid order data            | Order ID generated and saved | Enter result | Pending/Pass |
| View order                | Retrieve order by ID/user          | Correct order displayed      | Enter result | Pending/Pass |
| Update status             | Perform valid status change        | Status saved and shown       | Enter result | Pending/Pass |
| Reject invalid transition | Attempt prohibited change          | System blocks update         | Enter result | Pending/Pass |
| Access control            | Try protected action as wrong role | Unauthorized action denied   | Enter result | Pending/Pass |

## 4.4 Results

No actual test measurements or screenshots were supplied with the source template. Therefore, this report does not fabricate numerical performance values. The team should enter actual pass/fail results, response times and screenshots after testing the completed implementation.

## 4.5 Data Analysis & Engineering Judgment

Functional results should be analyzed by comparing expected and observed behaviour for each requirement. For response-time measurements, repeated trials can be averaged and the range can be recorded. Failed tests should be investigated through database logs, validation rules, network behaviour and user-interface logic. Engineering judgment should consider correctness, usability, reliability and security rather than only the number of features.

## 4.6 Comparison with Existing Solutions & Achievement of Objectives

| Objective                | Evidence                         | Achievement              |
|--------------------------|----------------------------------|--------------------------|
| Identify requirements    | Requirement specification        | Fully/Partially – verify |
| Design database/workflow | Schema and order-state flow      | Fully/Partially – verify |
| Implement tracking       | Tracking screen and status logic | Fully/Partially – verify |
| Test system              | Completed test cases             | Pending until measured   |

## 4.7 Strengths & Limitations

**Strengths:** clear order lifecycle; centralized records; role-oriented workflow; easy-to-understand tracking display; modular architecture.

**Limitations:** academic prototype scale; live GPS may not be included; real payment integration may be absent; production security and high-availability requirements are outside scope; delivery time estimates depend on external factors and should not be treated as guarantees.

## 4.8 Project Planning, Roles & Budget

| Item                     | Estimated Cost                    | Actual Cost  |
|--------------------------|-----------------------------------|--------------|
| Development software     | Free/open-source where applicable | Enter actual |
| Hosting/database         | Enter actual plan                 | Enter actual |
| Domain (if used)         | Enter actual                      | Enter actual |
| Testing/device resources | Enter actual                      | Enter actual |

# CHAPTER 5 CONCLUSION, FUTURE WORK AND REFLECTION

## 5.1 Conclusion

The Food Delivery Order Tracking System addresses the problem of limited visibility after a food order is placed. The proposed software organizes an order into a defined lifecycle and provides a centralized method for recording and displaying status. The design combines user interaction, database storage, role-based actions and tracking logic. Final claims about performance should be based on the team's completed functional tests.

## 5.2 Future Work

Future versions can include live GPS tracking, map visualization, push notifications, estimated delivery time calculation, restaurant POS integration, online payments, analytics, scalable cloud deployment and stronger security controls. A production system would also require load testing, privacy review, monitoring, backup and recovery procedures.

## 5.3 Professional Learning & Individual Reflection

**Dinesh:** The project develops experience in backend logic, database operations and order processing. The main learning is maintaining consistent records across multiple application actions. Future work would focus on stronger API validation and database optimization.

**Kiruthika:** The project develops skills in interface design, user interaction and integration testing. The main learning is that a tracking interface must communicate status clearly without confusing users. Future work would improve responsive design and accessibility.

**Akshaya:** The project develops system-level design, workflow modelling, documentation and presentation skills. A key engineering decision is to use a defined status lifecycle so that the order state remains understandable and consistent. If repeated, more user testing and measured response-time data would be collected before final evaluation.

## REFERENCES

Use a consistent recognised referencing style such as IEEE. Include only sources actually consulted by the team.

1. Documentation for the selected programming language/framework, official documentation, accessed September 2026.
2. Documentation for the selected database management system, official documentation, accessed September 2026.
3. Documentation for the selected web/API development tools, official documentation, accessed September 2026.
4. Relevant academic literature consulted on online food ordering, database-backed tracking systems, software engineering and user-interface design.
5. Any datasets, open-source libraries, websites or AI-assisted resources actually used in the implementation should be listed with complete bibliographic details.

## APPENDICES

### Appendix A – Detailed Design

Insert final architecture diagram, workflow diagram, database schema and screen designs.

### Appendix B – Database / Data Dictionary

Insert tables, fields, data types, keys and relationships used in the final system.

### Appendix C – Source Code / Repository Information

Insert the approved repository information or essential source-code excerpts.

#### **Appendix D – Screenshots / Implementation Evidence**

Insert login, menu, order creation, restaurant status, delivery status and customer tracking screenshots.

#### **Appendix E – Test Evidence**

Insert dated test cases, outputs, screenshots and defect/fix records.

#### **Appendix F – User Feedback**

Include feedback if collected.

#### **Appendix G – Project Meeting / Progress Records**

Include review notes, milestones and progress evidence.

#### **Appendix H – Individual Contribution Evidence**

Include task records, code contributions, design files and other evidence supporting the contribution statement.